Sample size and Power

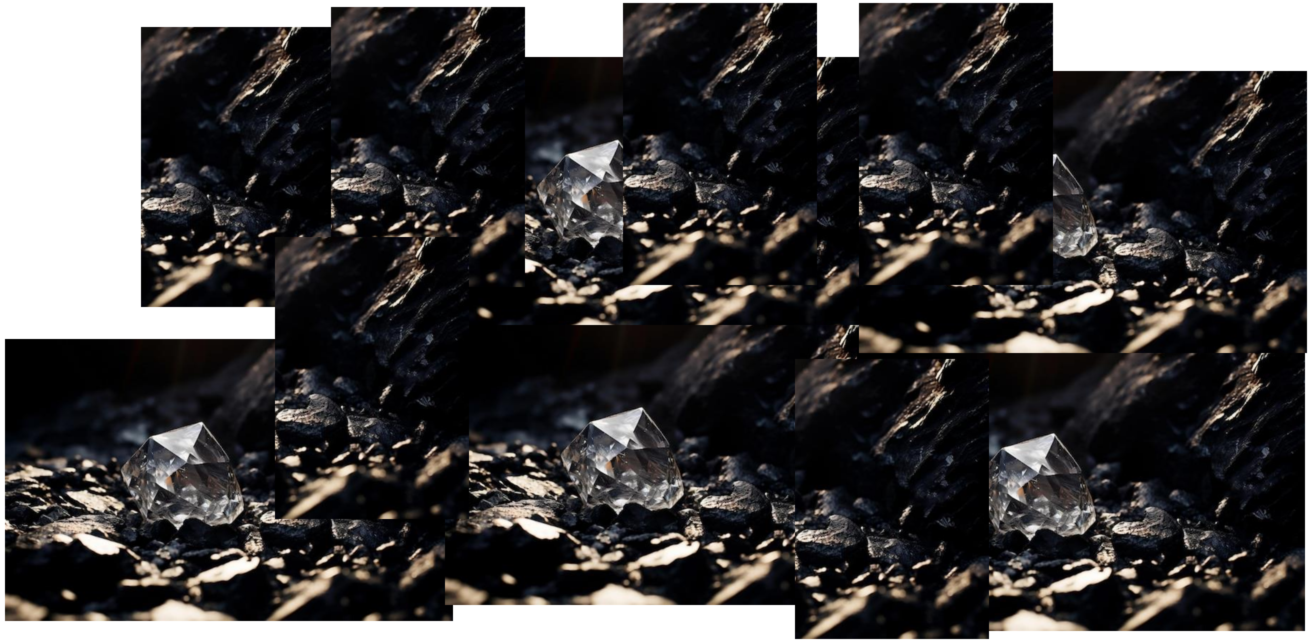

TREASURE HUNT
the quest for real value

The Diamond Hunt: Sample Size

Imagine you're digging through a pile of rocks to find diamonds…

Small sample size

Large sample size

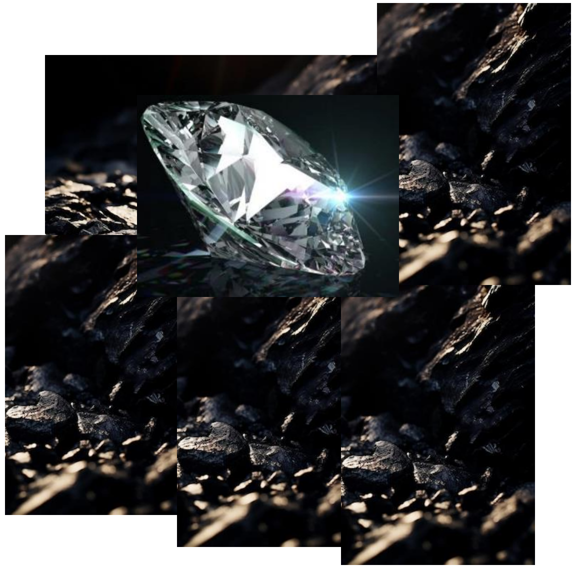


The Diamond Hunt: Effect Size

Imagine you’re digging through a pile of rocks to find diamonds…

Small effect size

Large effect size



The Diamond Hunt: Power

Imagine you're digging through a pile of rocks to find diamonds…

High power

Low power

The Diamond Hunt: Power

What determines power?

Low variability (noise)

High variability (noise)

The Diamond Hunt: Power

What determines power?

Low significance level: Only picking the flawless diamonds

High significance level: Picking all these options

The Diamond Hunt: Power

What determines power?

Not-so-good tools (study design)

Better tools (study design)

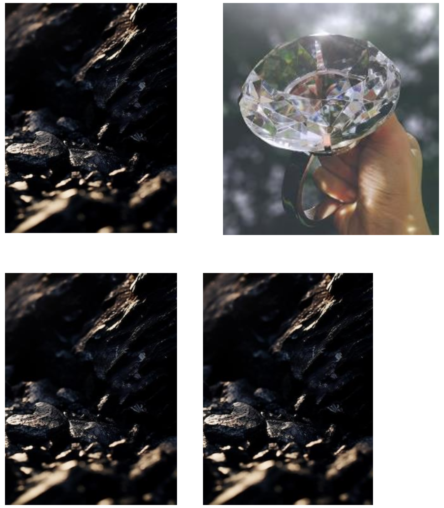
Hypothesis testing

- **Null Hypothesis (H_0)**: The default assumption that there is **no effect** or **no relationship** between the variables being tested.
- **Alternative Hypothesis (H_1)**: The opposite of the null hypothesis. It suggests that there **is an effect** or **a relationship** between the variables.
- Collecting evidence to test the Hypothesis
- **Reject** or **Fail to reject** the null hypothesis
- Just because you're testing doesn't mean you're always right!

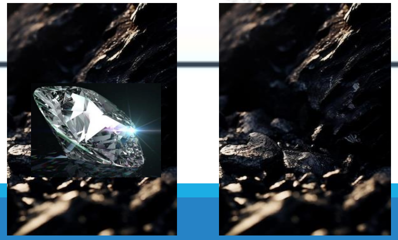
Errors in testing the null hypothesis

Rejecting the null hypothesis when it's actually true.

Measured / Perceived	Reality: True	Reality: False
True	Correct ☺	Type I False Positive
False	Type II False Negative	Correct ☺



Failing to reject the null hypothesis when it's false.



Review

What is statistical power?

- The probability of finding an effect if it exists (correctly rejecting the false null hypothesis)

What affects power?

- Sample size, effect size, significance level, noise/variability, study design

What are the potential errors in hypothesis testing?

- Type I (found an effect but it didn’t exist) or Type II (failing to find the effect when it existed)

Your turn! (Discussion Notebook)

Design a study to test whether positive reinforcement improves group performance in a collaborative task.

- Identify Your Hypotheses
- Define Key Variables
- Design your study and identify elements that effect power (try to maximize your power with a feasible design)
- Consider errors (what might lead to type I and type II errors?

Large scale replication attempts

LAB #8

Real world research practice issues (EC)

Research Group A conducted a longitudinal study, measuring the effects of a health program on physical and psychological health outcomes every month for a year. They claimed that physical health *caused* changes in psychological health and vice versa. I asked them how they determined the bidirectional causality between the two outcome variables. They said, "because this is a longitudinal study."

- What's the issue with this claim? When can we make a causal claim? What do we need to determine a causal relationship?

Real world research practice issues (EC)

Research Group B conducted an intervention experiment testing whether a food literacy educational program was effective in reducing the rate of food insecurity. They measured participants' food insecurity levels at baseline, 6-month follow-up, and 12-month follow-up. They found that, "although there was no change for the intervention group from baseline to the 6-month follow-up, the food insecurity rate dropped to 0% at the 12-month follow-up. Our intervention program is working well!" However, they lost almost half of the participants between the 6-month and 12-month follow-ups. The researchers did not provide details on why participants dropped out or if the dropout was random. They also did not mention if any sensitivity analyses were conducted to address the missing data.

- Do you think their conclusion is problematic? Why?

Real world research practice issues (EC)

Research Group C did an observational study on moderate-to-vigorous physical activity (MVPA). They measured participants' total MVPA each day (in minutes) as a continuous variable. However, in their presentation, they dichotomized the continuous variable into two levels: high MVPA and low MVPA. When I asked why, they said, "We used the continuous variable at first, but there was no significant effect; we saw a significant effect after dichotomizing it."

- Do you think this is an acceptable practice? Why?

Replicability (Discussion Notebook)

How does social media impact self-esteem?

Methods

Participants: The study involved a sample of 150 university students aged 18-24, recruited from introductory psychology courses.

Procedure: Participants completed an online survey at the beginning and end of a four-week period. The survey included questions about their social media usage (time spent, platforms used) and self-esteem levels.

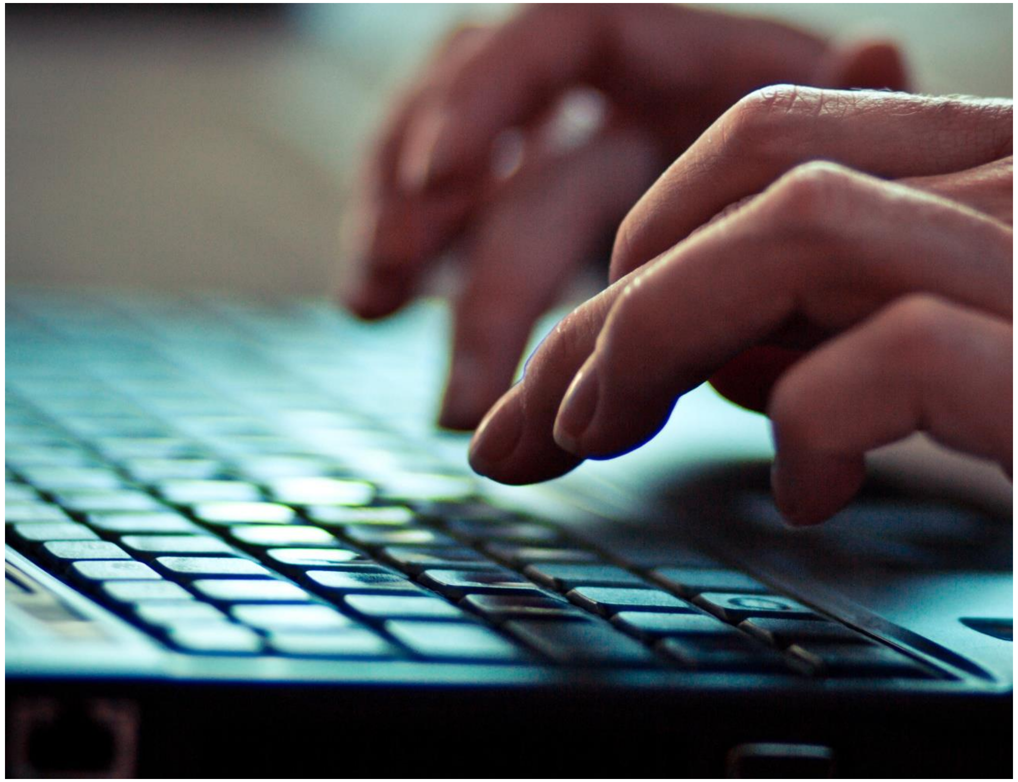
Materials: The Rosenberg Self-Esteem Scale (RSES) was used to measure self-esteem. Social media usage logs were self-reported by participants.

Analysis: Data were analyzed using t-tests to compare pre- and post-study self-esteem scores, and correlation analyses were performed to examine the relationship between social media usage and self-esteem.

Answer the following questions (DN)

1. What details are missing from the methods section that would make a direct replication challenging?
2. What more would you need to replicate the study exactly as the authors did it?
3. As it is written now, do you have confidence in whether the authors' findings truly captured what they claimed to be studying?

How replications are done



Work with original authors

How replications are done

Get all protocols and details possible about the design, measures, analyses, etc.

How replications are done

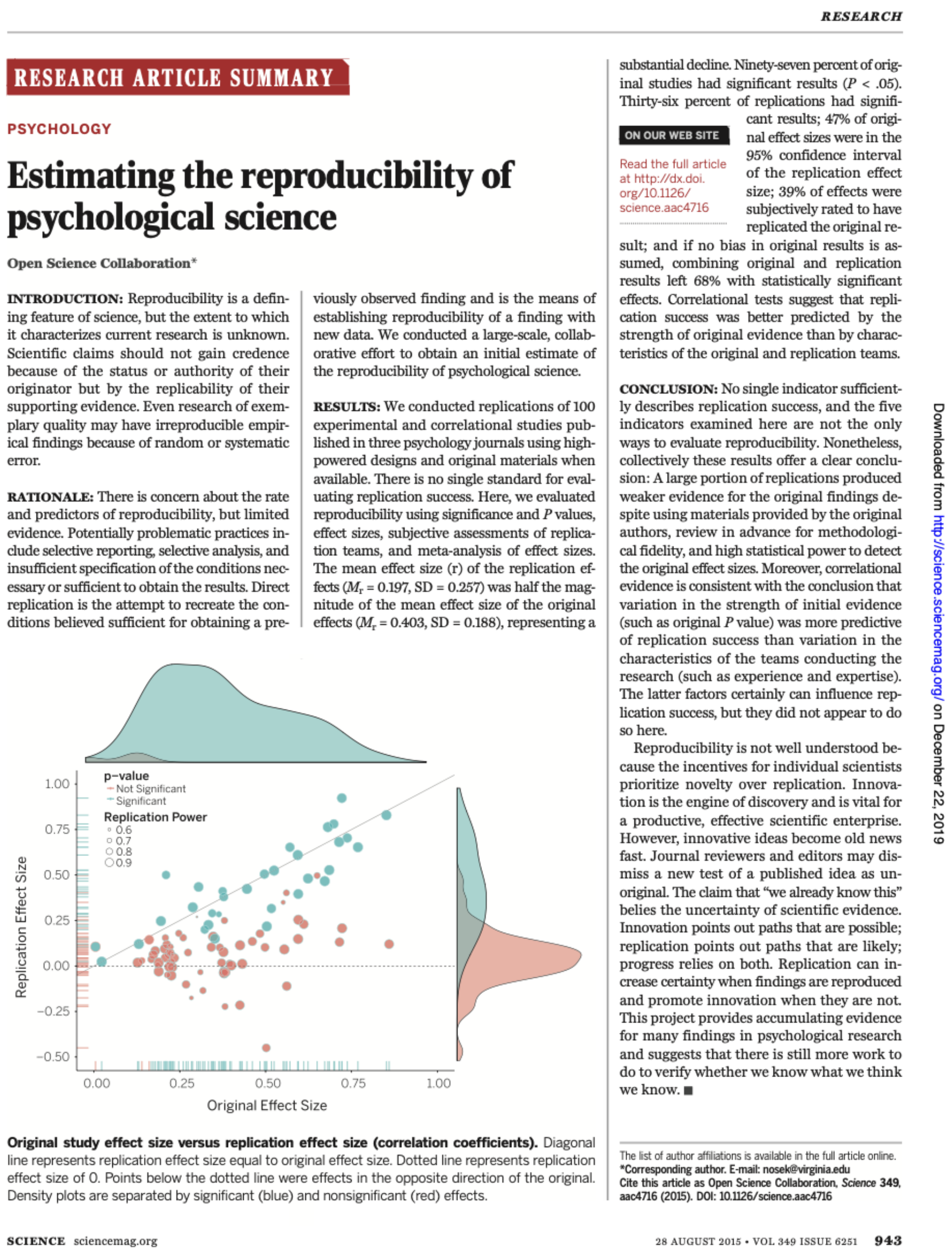
Register the study

OSF PREREGISTRATION

PREREGISTERED

How replications are done

- Run the study
- Repeat
- Make all data/materials publicly available
- Publish



RESEARCH ARTICLE SUMMARY

PSYCHOLOGY

Estimating the reproducibility of psychological science

Open Science Collaboration*

INTRODUCTION: Reproducibility is a defining feature of science, but the extent to which it characterizes current research is unknown. Scientific claims should not gain credence because of the status or authority of their originator but by the replicability of their supporting evidence. Even research of exemplary quality may have irreproducible empirical findings because of random or systematic error.

RATIONALE: There is concern about the rate and predictors of reproducibility, but limited evidence. Potentially problematic practices include selective reporting, selective analysis, and insufficient specification of the conditions necessary or sufficient to obtain the results. Direct replication is the attempt to recreate the conditions believed sufficient for obtaining a previously observed finding and is the means of establishing reproducibility of a finding with new data. We conducted a large-scale, collaborative effort to obtain an initial estimate of the reproducibility of psychological science.

RESULTS: We conducted replications of 100 experimental and correlational studies published in three psychology journals using high-powered designs and original materials when available. There is no single standard for evaluating replication success. Here, we evaluated reproducibility using significance and P values, effect sizes, subjective assessments of replication teams, and meta-analysis of effect sizes. The mean effect size (r) of the replication effects (M_r = 0.197, SD = 0.257) was half the magnitude of the mean effect size of the original effects (M_r = 0.403, SD = 0.188), representing a substantial decline. Ninety-seven percent of original studies had significant results (P < .05). Thirty-six percent of replications had significant results; 47% of original effect sizes were in the 95% confidence interval of the replication effect size; 39% of effects were subjectively rated to have replicated the original result; and if no bias in original results is assumed, combining original and replication results left 68% with statistically significant effects. Correlational tests suggest that replication success was better predicted by the strength of original evidence than by characteristics of the original and replication teams.

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Read the full article at http://dx.doi.org/10.1126/science.aac4716

CONCLUSION: No single indicator sufficiently describes replication success, and the five indicators examined here are not the only ways to evaluate reproducibility. Nonetheless, collectively these results offer a clear conclusion: A large portion of replications produced weaker evidence for the original findings despite using materials provided by the original authors, review in advance for methodological fidelity, and high statistical power to detect the original effect sizes. Moreover, correlational evidence is consistent with the conclusion that variation in the strength of initial evidence (such as original P value) was more predictive of replication success than variation in the characteristics of the teams conducting the research (such as experience and expertise). The latter factors certainly can influence replication success, but they did not appear to do so here.

Reproducibility is not well understood because the incentives for individual scientists prioritize novelty over replication. Innovation is the engine of discovery and is vital for a productive, effective scientific enterprise. However, innovative ideas become old news fast. Journal reviewers and editors may dismiss a new test of a published idea as unoriginal. The claim that "we already know this" belies the uncertainty of scientific evidence. Innovation points out paths that are possible; replication points out paths that are likely; progress relies on both. Replication can increase certainty when findings are reproduced and promote innovation when they are not. This project provides accumulating evidence for many findings in psychological research and suggests that there is still more work to do to verify whether we know what we think we know. ■

Original study effect size versus replication effect size (correlation coefficients). Diagonal line represents replication effect size equal to original effect size. Dotted line represents replication effect size of 0. Points below the dotted line were effects in the opposite direction of the original. Density plots are separated by significant (blue) and nonsignificant (red) effects.

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Estimating the reproducibility of psychological science (DN)

What are some of the key factors that the Open Science Collaboration (2015) identified as **contributors to the replication crisis** in psychological science, and how can researchers **address these factors** to improve the reliability of scientific findings?

McShane et al., 2019 (DN)

- What are the author’s arguments for why large-scale replication attempts are needed?
- What do the authors mean by how heterogeneity is a factor in these replications?

Large-Scale replication groups

- Psychological Science Accelerator
- Association for Psychological Science
- StudySwap
- Collaborative Replications and Education Project (CREP)
- Reproducibility project: Psychology